

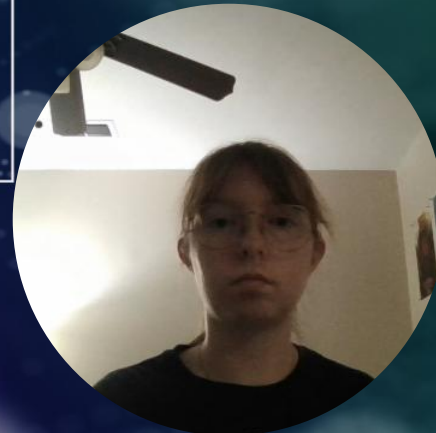


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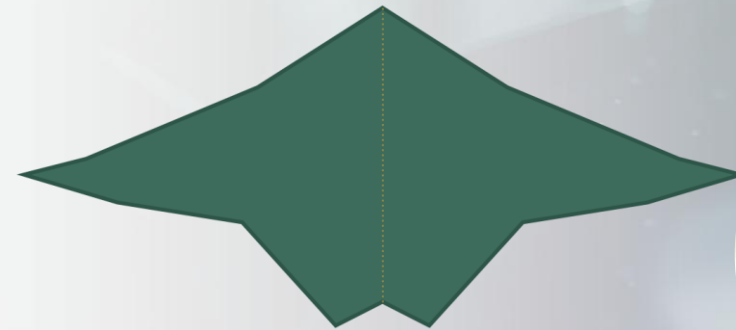
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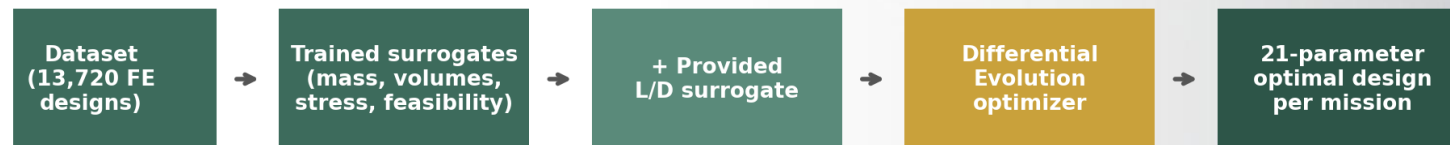
AI-Driven BWB Inverse Design

- Objective: automated inverse design → optimal BWB planform + internal structure
 - Inputs: target L/D, minimum payload/fuel volume, flight condition
 - Outputs: **21** design parameters (**10** planform + **11** structural)
- Constraint: no external CFD/FEA — built on the provided **13,720-design** dataset + L/D surrogate



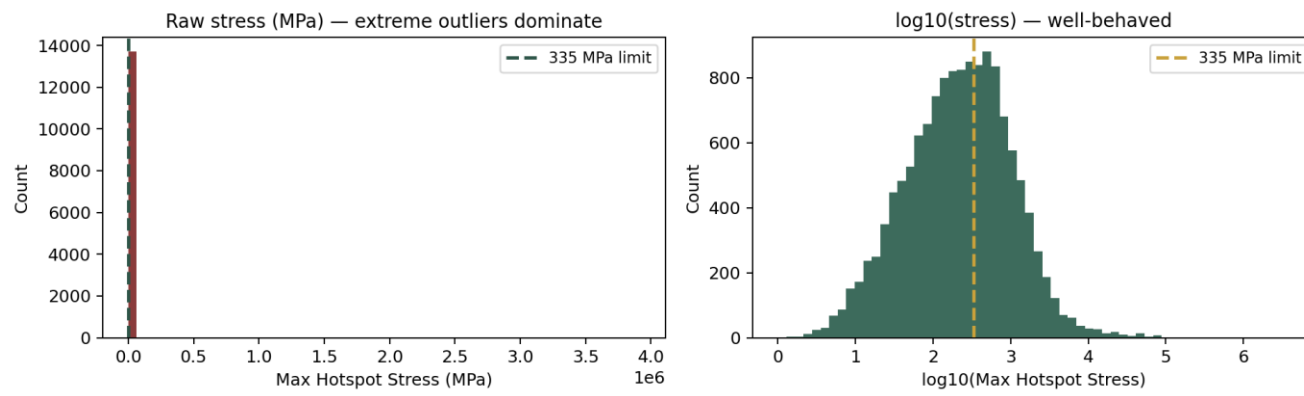
Approach: Surrogate-Driven Optimization

- Trained gradient-boosted tree surrogates on the provided dataset:
 - Mass, payload volume, fuel volume — $R^2 > 0.98$
 - Stress — trained in log-space due to extreme right-skew
- Used the organizers' provided L/D surrogate directly for aerodynamics
- Differential evolution (gradient-free, population-based) searches the full 21-variable space to minimize mass subject to mission targets



Handling the Hard Stress Constraint

- Raw stress is extremely right-skewed (median ~228 MPa, outliers past **1M MPa**) — a direct regressor performed **worse than guessing the mean**
- Fix: trained on $\log_{10}(\text{stress})$ instead — the **335 MPa** limit sits in the dense part of that distribution
- Hidden ground-truth validation → feasibility requires **BOTH** checks to pass:
 - Classifier confidence ≥ 0.92
 - Stress regressor estimate $\leq 300 \text{ MPa}$ (margin below the 335 MPa limit)



Results Across the 3 Mission Cases

- All three optimized designs are feasible under the conservative gate
- Mass scales sensibly with mission demand — highest-capacity mission produces the heaviest structure
- Fuel volume is the hardest target to hit consistently across missions



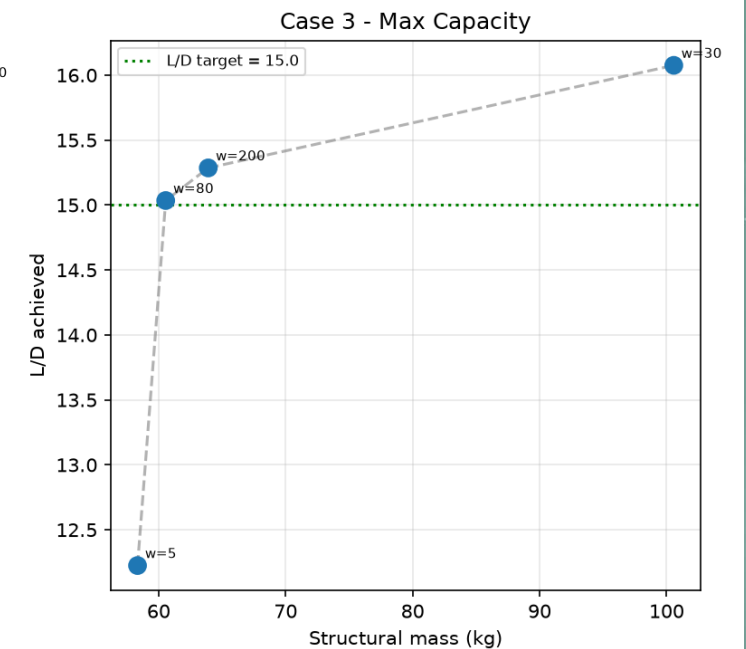
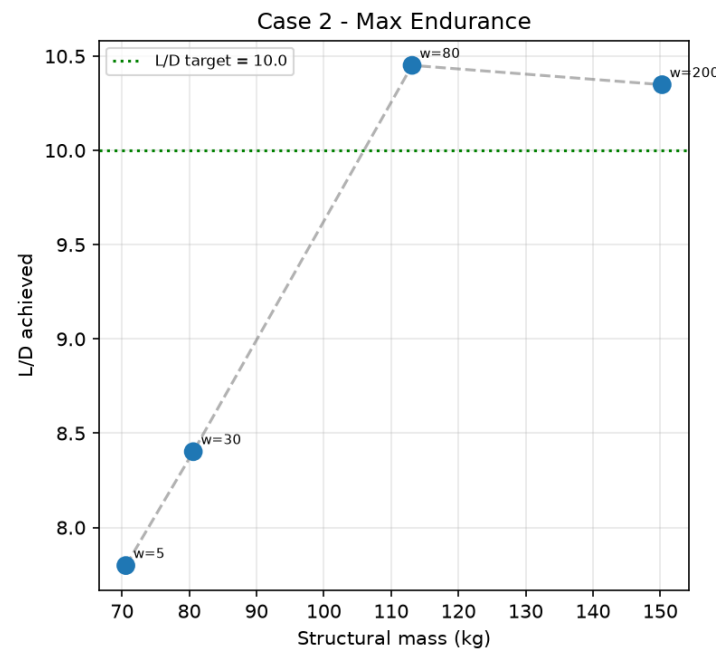
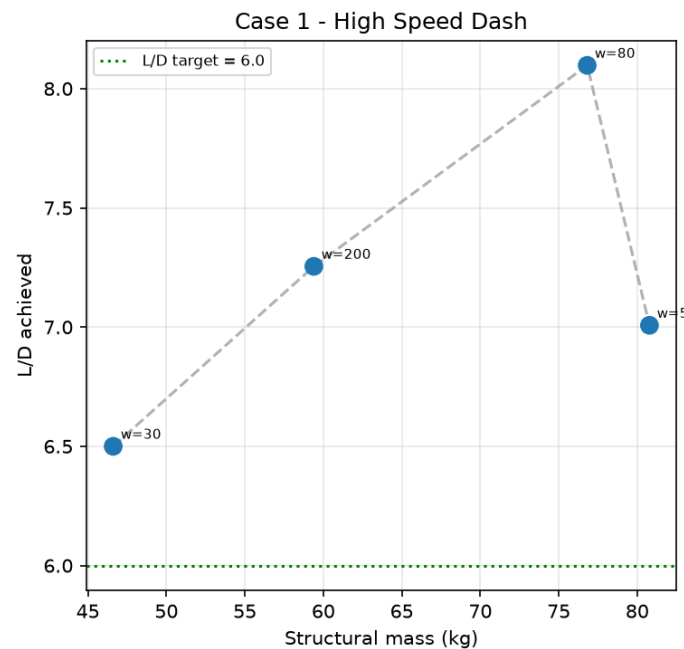
Mission	Mass (kg)	L/D ach.	L/D target	Payload ach. (m3)	Payload min (m3)	Fuel ach. (m3)	Fuel min (m3)	Feasible prob
High Speed Dash	33.1	7.61	6.0	0.772	0.75	0.435	0.45	0.93
Max Endurance	50.2	10.05	10.0	0.925	0.80	0.456	0.45	0.95
Max Capacity	54.8	14.80	15.0	0.480	1.00	0.335	0.65	0.92





Multidisciplinary Trade-off: Mass vs. L/D

- Re-ran the optimizer across a sweep of L/D-priority weights per mission
- Traces an approximate Pareto front between structural mass and aerodynamic efficiency — not just one answer



Takeaways

- Surrogate-driven inverse design pipeline respecting the hard stress constraint conservatively
- Explicit mass / L-D trade-off analysis, not just a single optimum
- With more time: higher-resolution Pareto sweep, ensemble surrogates for extra stress-prediction robustness

